

A Comparative Analysis of Large Language Model(LLM) and Human-Generated Text Using Natural Language Processing

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1. Introduction

Can we still tell apart human and machine-generated text?

The line between Human and AI is fading fast

- Modern Large Language Models like GPT and Gemini now produce content nearly indistinguishable from human writing.
- These models can convincingly mimic human style, tone, and reasoning in a wide range of contexts.
- This remarkable advancement raises a critical challenge: ensuring the reliability and authenticity of digital content.
- Without clear classification, misinformation, AI-generated spam, and academic dishonesty may spread unchecked.
- Such risks affect vital areas like education, journalism, and cybersecurity, where trust is essential.
- Therefore, distinguishing between human and machine-generated text is no longer optional, but necessary

Classifying text origin is essential to ensure trust, safety, and ethical AI use

1. Introduction

Related work about text classification Based Natural Language Processing

D. Mamata, K. Selvakumar , P.J.A. Alphonse, A **Comparative** Study on **TF-IDF** feature Weighting Method and its Analysis using Unstructured Dataset, arXiv:2308.04037 (2023)

Kim, SW., Gil, JM. Research paper **classification systems** based on **TF-IDF** and **LDA** schemes. Hum. Cent. Comput. Inf. Sci. 9, 30 (2019)

Essa, E., Omar, K. & Alqahtani, A. **Fake news detection** based on a hybrid BERT and **LightGBM** models. Complex Intell. Syst. 9, 6581–6592 (2023).

M. H. Ibnath, M. Sheikh, M. Sheikh and J. F. Shormy, **Enhanced Suicidal Ideation Detection** via **Text Classification** and Ensemble Methods, 2025 International Conference on Electrical, pp. 1-6 (2025).

Voskergian, D., Jayousi, R. & Yousef, M. Topic selection for **text classification** using ensemble **topic modeling** with grouping, scoring, and modeling approach. Sci Rep 14, 23516 (2024).

K. H. Lee, **Natural Language Processing Based Comparative Analysis of Research, News, and Complaints Data** in the Cadastre Domain, ksc vol.41, no.1, pp.83-97 (2025).

Essa, E., Omar, K. & Alqahtani, A. **Fake news detection** based on a **hybrid BERT** and **LightGBM** models. Complex Intell. Syst. 9, 6581–6592 (2023).

2. Data & Methodology

TF-IDF

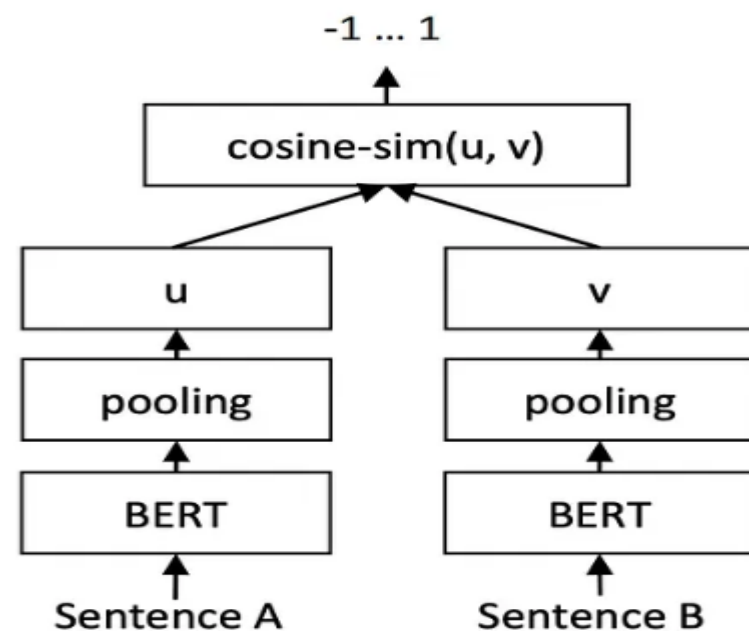
$$TF(t, d) = \frac{(\text{Number of occurrences of term } t \text{ in document } d)}{(\text{Total number of terms in the document } d)}$$

$$IDF(t, D) = \log_e \frac{(\text{Total number of documents in the corpus})}{(\text{Number of documents with term } t \text{ in them})}$$

$$TF-IDF(t, d, D) = TF(t, d) \times IDF(t, D)$$

Term frequency (TF) multiplied by inverse document frequency (IDF) to represent how important a word is in a particular document

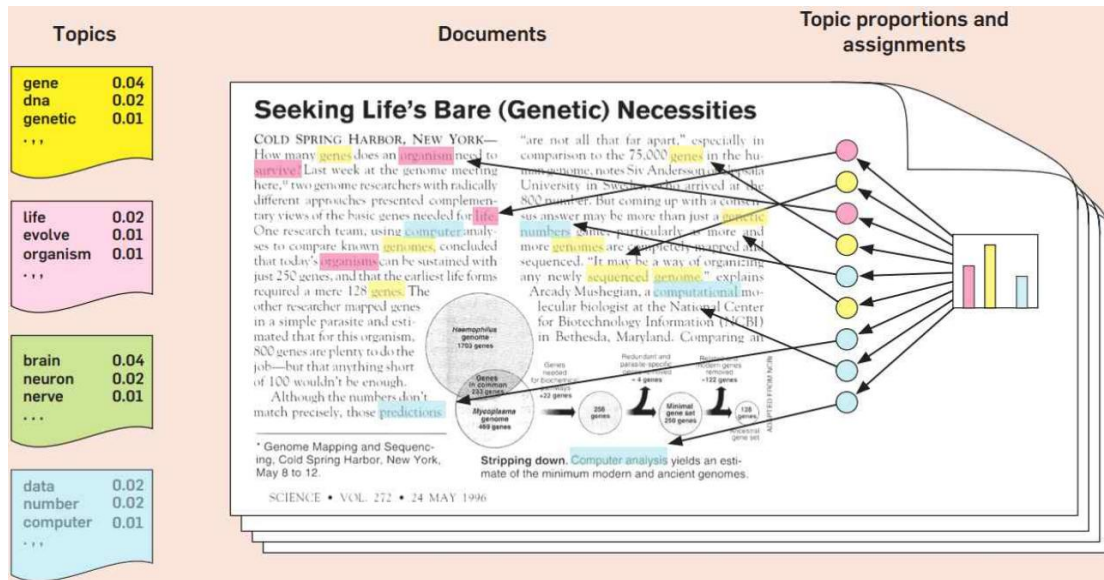
Sentence Bert & Cosine Similarity



SBERT architecture at inference, for example, to compute similarity scores. This architecture is also used with the regression objective function

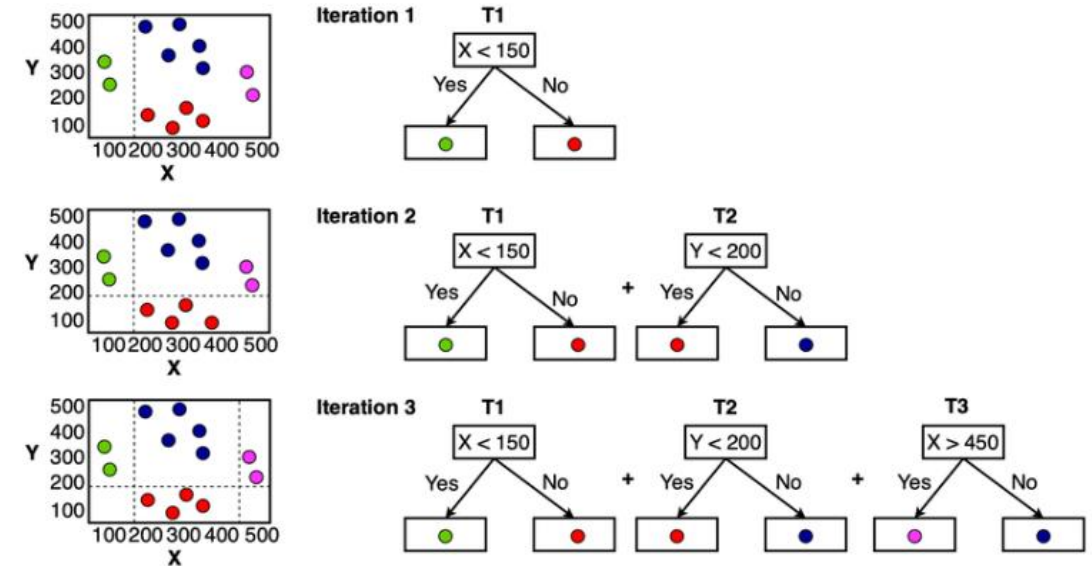
2. Data & Methodology

Topic Modeling



LDA techniques infer unobserved variables from the observed variables in the data. Therefore, it can effectively deal with textual data by identifying, separating, and integrating keywords with the same meaning to discover the topic of the text

LightGBM



Algorithms use an iterative process called boosting, which sequentially improves the base learner to build a resilient predictive model. By leveraging gradient descent to optimize the loss function during each iteration, increasingly accurate models can be developed

2. Data & Methodology

Data Collection : Building a Diverse Corpus

Human Text

Quora

A Place to share Knowledge and
Better understand the world



~2800 responses collected from Quora
(up to 30 answers per question)

LLM Text



3000 responses from each of the four LLMs
(30 answers x 100 questions)

Curated a set of 100 unique questions spanning a diverse range of topics to ensure the generality

2. Data & Methodology

Data Collection : Building a Diverse Corpus

Prompt Typology

Statistical Analysis

Minimal Guidance

- Uses minimal, neutral instructions to capture the LLM's default response style

Objective Response

- Directs the LLM to prioritize factual, objective information, testing its ability

Experience Sharing

- Aims to elicit subjective, experience-based narratives and opinions, assessing the LLM's capacity for a personal

Analytical Response

- Encourages detailed, analytical answers that push the LLM to explore topics comprehensively

TF-IDF, SBERT, Classification

Core Informative & Analytical Set

- Assesses versatile LLM responses by combining Minimal, Analytical, objective.
- Subjective experience prompts were excluded to first establish a standardized base-Line for core information processing capabilities.
- A total of 100 questions were curated across multiple topics, with 30 responses per question sourced from four LLMs and human contributors. While LLM responses were consistently sampled to provide 30 answers per question, the number of human responses varied, and some questions had fewer than 30 human answers available.

2. Data & Methodology

Use NLP techniques to distinguish between human and LLM generated text

Step.1 Data Collection and Processing

- Curated a dataset consisting of 100 questions and 30 diverse answers, standardized by NLTK preprocessing for uniform token count

Step.2 Part of Speech(POS)

- stylometry and Part-of-Speech (POS) analysis are utilized to measure the fundamental structure and grammatical characteristics of the text

Step.3 TF-IDF

- lexical signature analysis via TF-IDF identifies unique and significant keywords associated with each author

Step.4 Similarity analysis - Sentence-Bert

- semantic embeddings employing Sentence-BERT (SBERT) capture contextual meanings and semantic similarities within texts

Step.5 Hierarchical LDA-Based Subtopic Modeling

- topic modeling using Latent Dirichlet Allocation (LDA) explores hidden topics and underlying themes present in the texts

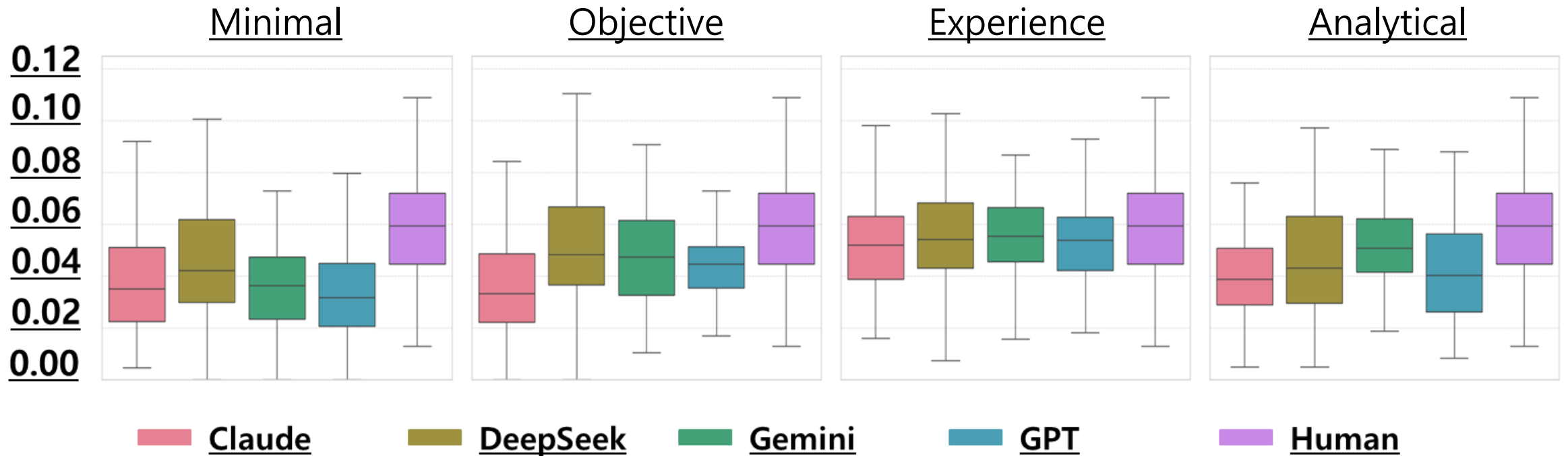
Step.6 Classification - LightGBM

- these derived features are integrated into a classification model trained with LightGBM to predict whether a text was authored by a human or a specific LLM

3. Syntactic Fingerprinting

How They Speak (1) – Who writes “me” a lot, and what will they talk about?

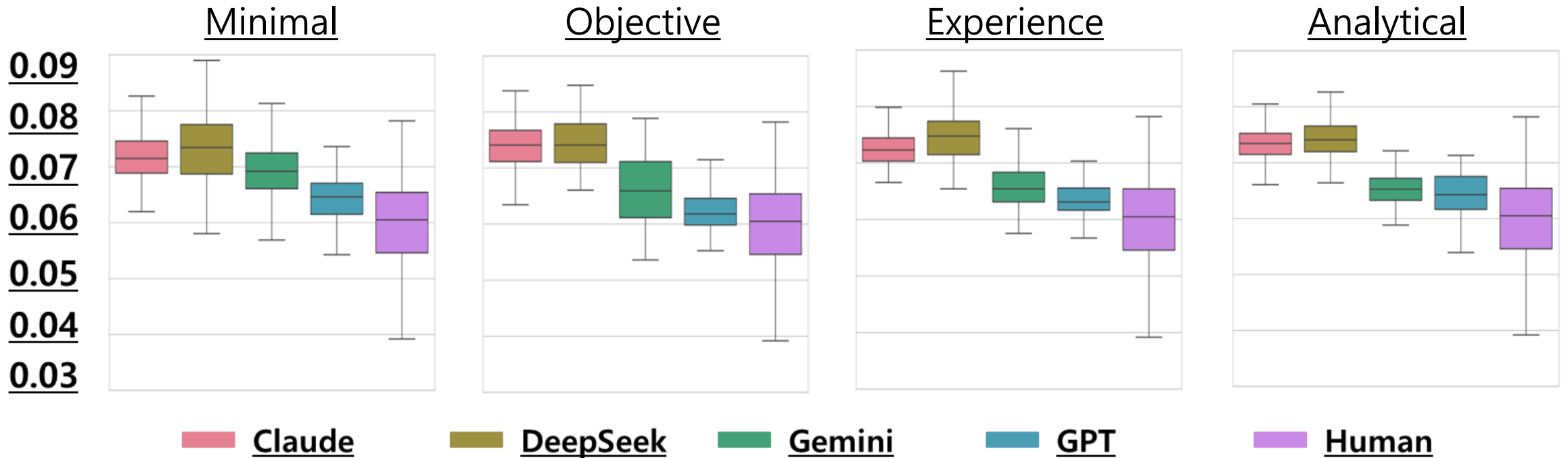
1. The average usage proportion is statistically higher in human responses
2. This suggests a strong tendency for humans to directly express their personal experiences, opinions, and emotions



3. Syntactic Fingerprinting

How They Speak (2) - How Diverse is Their Word Choice?

1. DeepSeek (avg. 0.75) and Claude (avg. 0.73) exhibit relatively high lexical density, indicating efficient use of diverse words relative to their answer length
2. In contrast, GPT (avg. 0.64) and Human (avg. 0.61) show a somewhat lower lexical density



4. Topical & Lexical Signatures

TF-IDF – Which word they choose ?

1. TF-IDF identifies uniquely important keywords by measuring how often a word appears in a document versus how rare it is across all documents
2. This lexical gap indicates a fundamental difference : LLMs function as generalist information processors, while human act as specialists of their own experience

$$TF - IDF = TF * IDF$$

Top-10	Human		GPT		Gemini		DeepSeek		Claude	
	Word	Score	Word	Score	Word	Score	Word	Score	Word	Score
1	One	0.2570	Like	0.3072	Like	0.3796	Like	0.3088	Like	0.2530
2	Time	0.2424	One	0.2373	Think	0.2711	One	0.2925	People	0.2247
3	Would	0.2235	Time	0.1840	Thing	0.2285	People	0.2696	I've	0.2189
4	People	0.1879	People	0.1837	One	0.1966	Even	0.2426	Think	0.2042
5	Like	0.1807	Think	0.1768	People	0.1805	Way	0.1815	Something	0.1804
6	Get	0.1582	Might	0.1735	Even	0.1754	Thing	0.1621	One	0.1750
7	Day	0.1550	Thing	0.1595	Okay	0.1751	Time	0.1506	Actually	0.1681
8	Year	0.1521	I've	0.1563	Really	0.1610	Make	0.1402	Thing	0.1529
9	Life	0.1457	Even	0.1517	Know	0.1607	Something	0.1389	Time	0.1517
10	dont	0.1381	Life	0.1473	something	0.1361	dont	0.1331	someone	0.1344

4. Topical & Lexical Signatures

Sentence BERT – How similar is their context ?

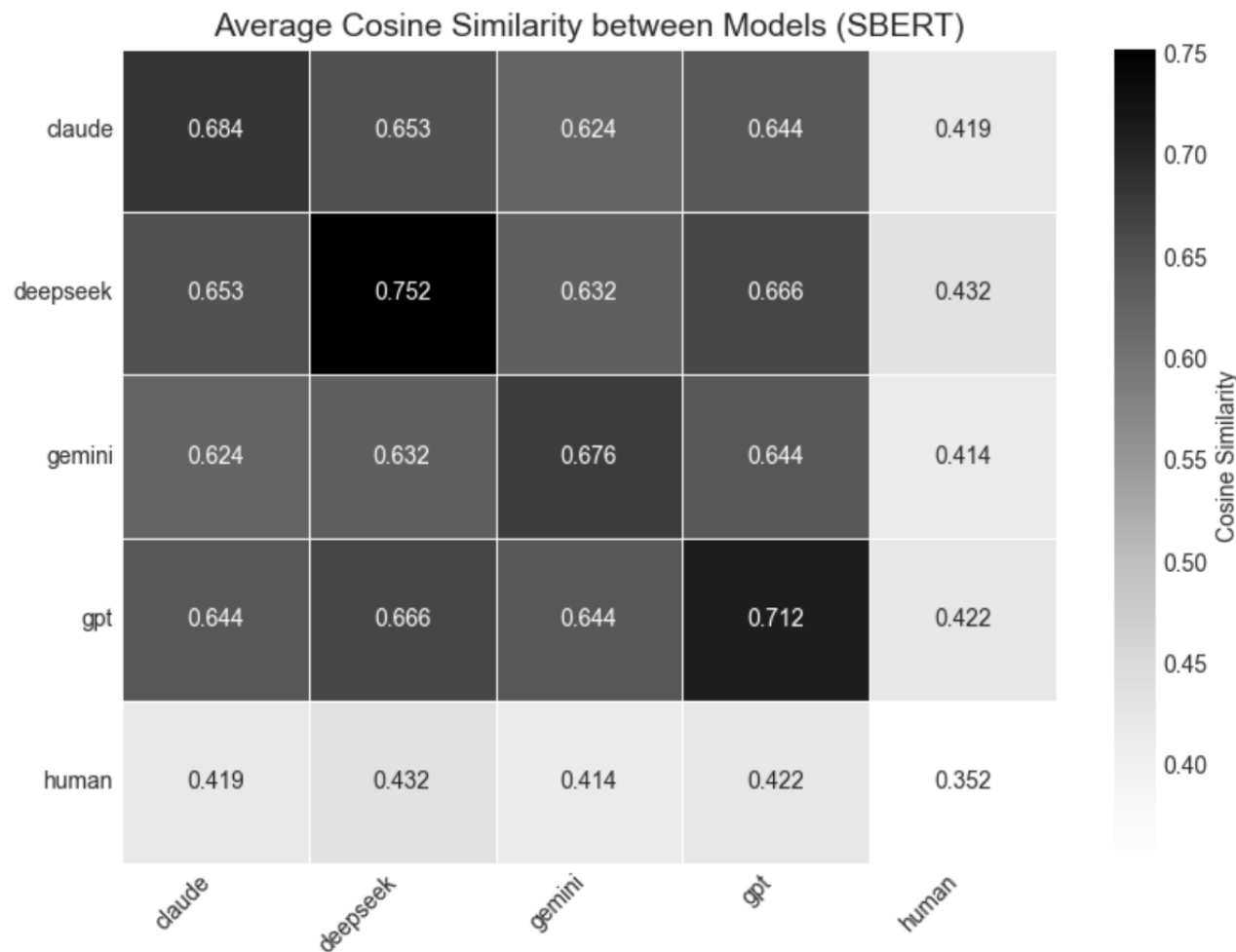
1. LLMs show high semantic similarity :

They consistently provide answers with similar meanings, reflecting a convergent thought process

2. Humans show high semantic

diversity : Their responses are distinct from LLMs and highly varied among themselves, reflecting unique individual perspectives

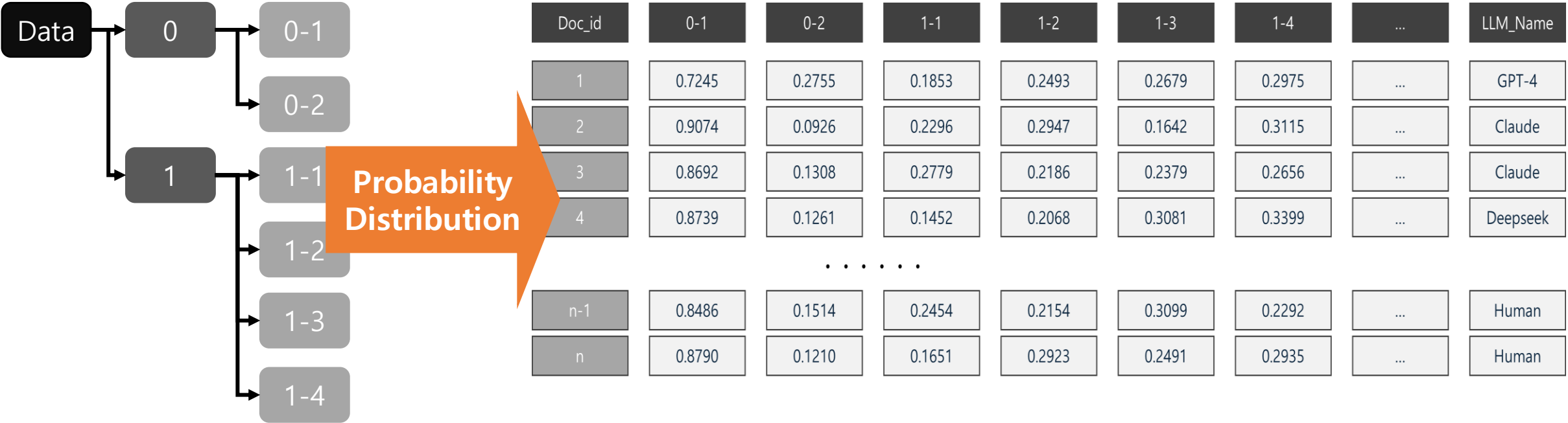
$$similarity = \cos(\theta) = \frac{A \cdot B}{\|A\| \|B\|} = \frac{\sum_{i=1}^n A_i \times B_i}{\sqrt{\sum_{i=1}^n (A_i)^2} \times \sqrt{\sum_{i=1}^n (B_i)^2}}$$



5. Classification

Hierarchical LDA – Based Subtopic Modeling

- 1. To capture nuanced thematic structure, each text was transformed into a vector of subtopic probabilities using a hierarchical LDA approach
- 2. This thematic feature vector served as the sole input for our classifier, allowing us to predict authorship based purely on differences in topic composition

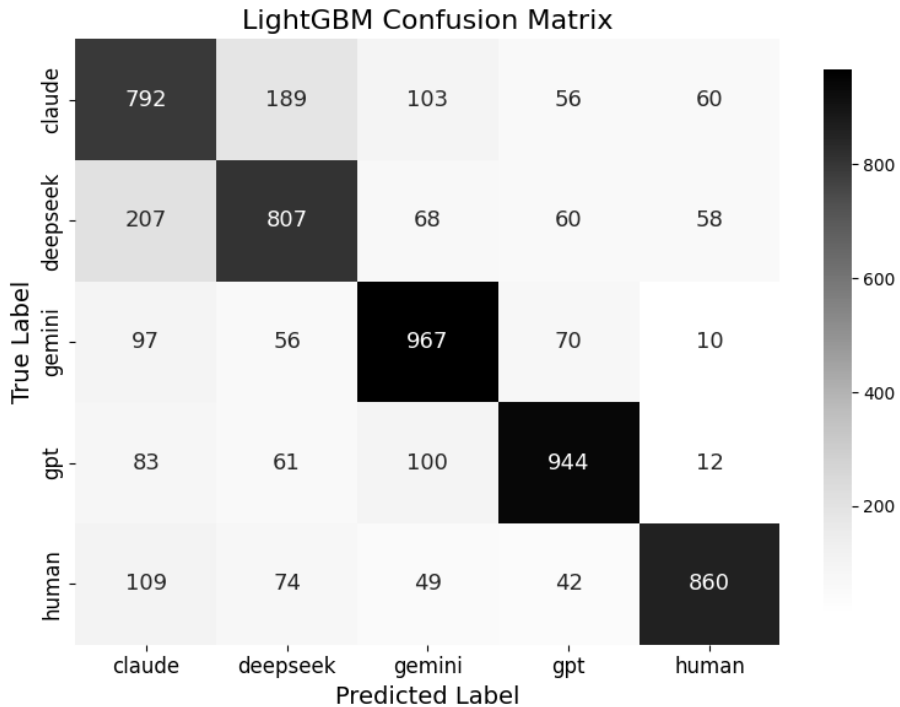


5. Classification

LightGBM Based Classification

- 1. Our LightGBM model achieved an overall accuracy of 73.6% using only subtopic features, proving that thematic structure is a powerful differentiator
- 2. The model classified human responses most accurately(F1-score : 0.805) and revealed distinct patterns among LLMs, with Gemini and GPT being the most recognizable

	precision	recall	f1-score	support
claude	0.614907	0.66	0.636656	1200
deepseek	0.679865	0.6725	0.676163	1200
gemini	0.75136	0.805833	0.777644	1200
gpt	0.805461	0.786667	0.795953	1200
human	0.86	0.758377	0.805998	1134
accuracy	0.736434			
Macro avg	0.742319	0.736675	0.738483	5934
Weighted avg	0.74101	0.736434	0.737732	5934



6. How to use LLM about RIMES ?

Related work about Materials Science using LLM

Kumar, Kumar, **Generative AI** in **Materials Science**: Accelerating Discovery Through Inverse Design (May 12, 2025). Available at SSRN:.

Henry W. et al, Monte Carlo Thought Search: **Large Language Model** Querying for Complex Scientific Reasoning in **Catalyst Design**, The 2023 Conference on Empirical Methods in Natural Language Processing

Zi-Yi Chen et al, MatChat: A **large language model** and application **service platform for materials science** 2023 Chinese Phys. B 32 118104

Z. Pei, J. Yin, J. Zhang, **Language models** for materials discovery and sustainability: Progress, challenges, and opportunities, Progress in **Materials Science**, Volume 154, 2025

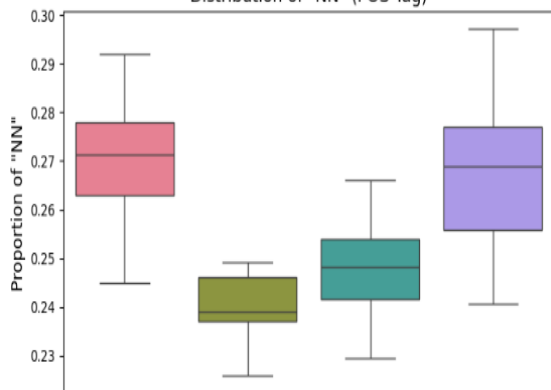
6. How to use LLM about RIMES ?

POS analysis

1. 내용 요약하기
2. 두줄로~ 위에 형식들 참고

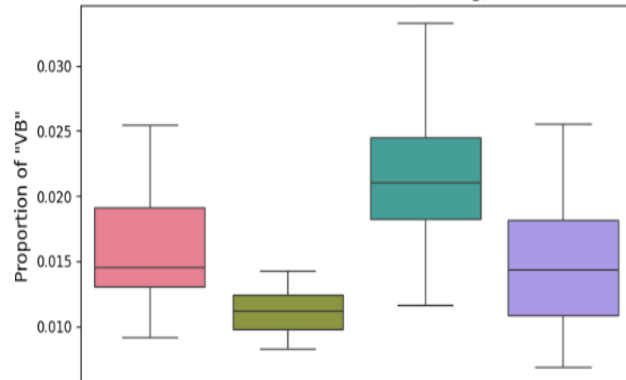
무슨 분석임?

Distribution of "NN" (POS Tag)



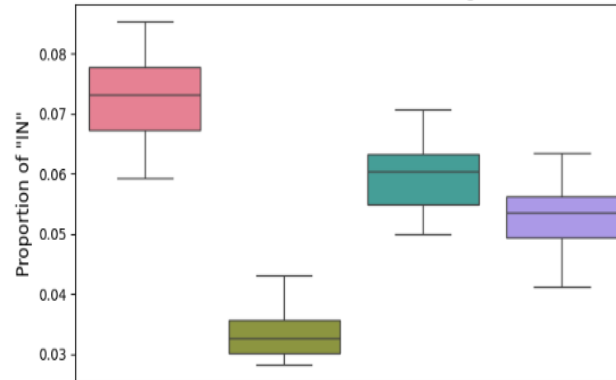
무슨 분석임?

Distribution of "VB" (POS Tag)



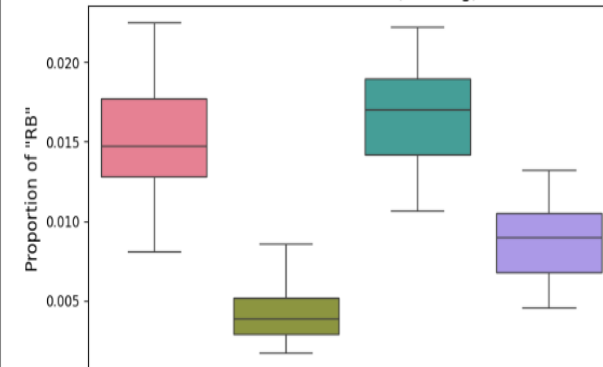
무슨 분석임?

Distribution of "IN" (POS Tag)



무슨 분석임?

Distribution of "RB" (POS Tag)



 Claude

 DeepSeek

 Gemini

 GPT

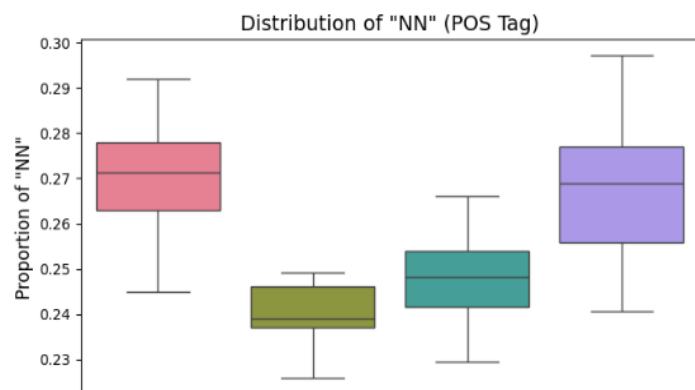
 Human

6. How to use LLM about RIMES ?

POS analysis

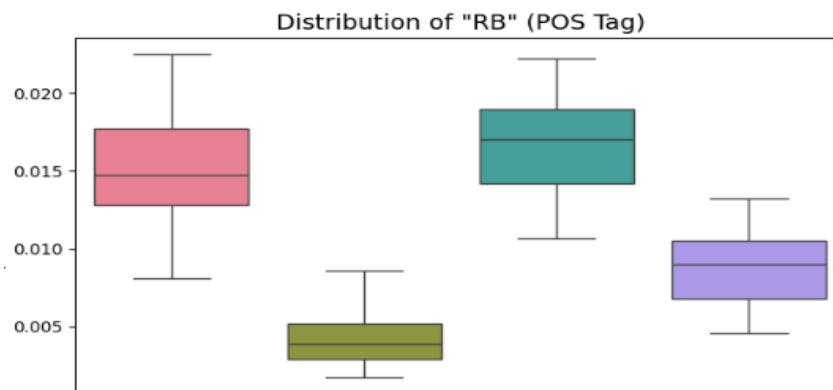
1. 내용 요약하기
2. 두줄로~ 위에 형식들 참고

무슨 분석임?



 Claude

무슨 분석임?

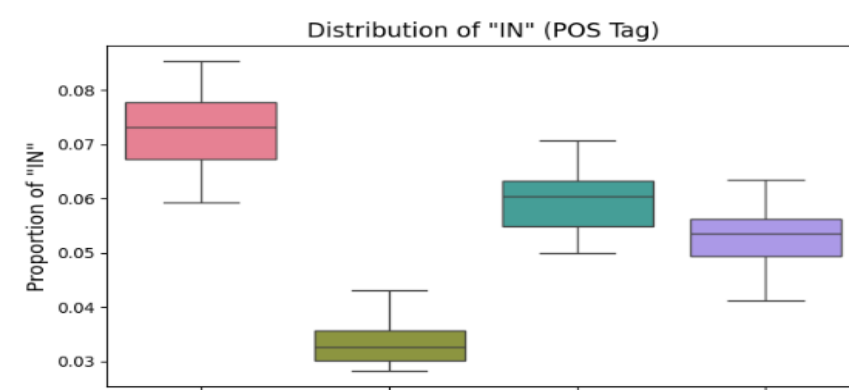


 DeepSeek

 Gemini

 GPT

무슨 분석임?



 Human

6. How to use LLM about RIMES ?

TF-IDF

- 1. Each LLM highlights different focus areas: GPT is descriptive, DeepSeek is process-oriented, Gemini emphasizes materials, and Claude leans toward principles
- 2. These patterns reflect their distinct strengths in technical explanation, practicality, materials, and theory

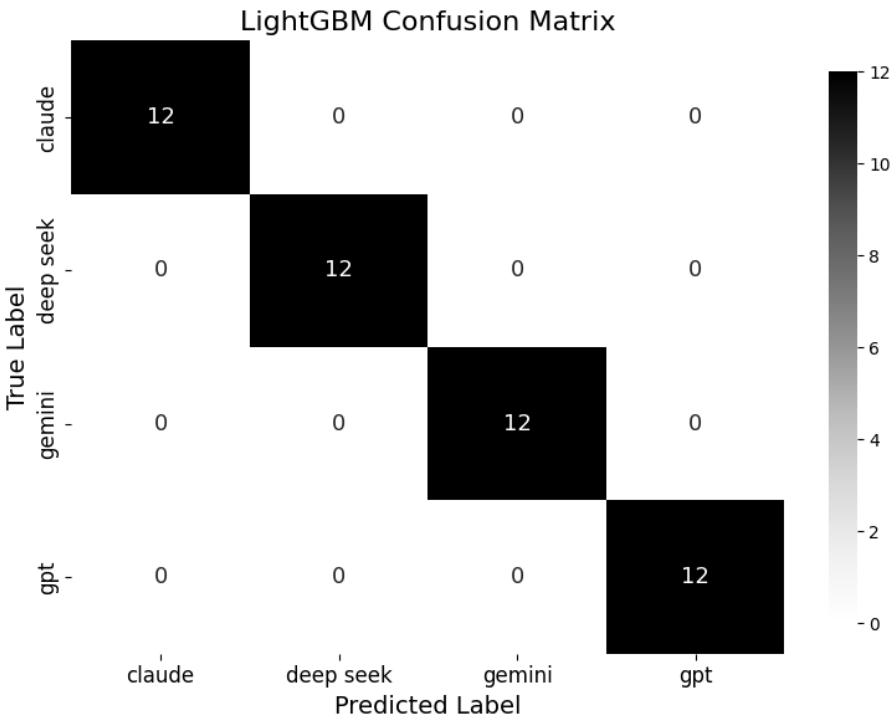
Top-10	GPT		Gemini		DeepSeek		Claude	
	Word	Score	Word	Score	Word	Score	Word	Score
1	Description	0.1369	Eg	0.1690	Advantage	0.2006	Core	0.1119
2	Advantage	0.0967	Sulfur	0.1188	Process	0.1427	Core Principle	0.0991
3	Core	0.0899	Carbon	0.1122	Hydrothermal	0.0827	Principle	0.0887
4	Principle	0.0847	Principle	0.0938	Hso	0.0773	Common synthesis	0.0884
5	Mos	0.0662	Description	0.0916	Mos	0.0725	Advantage	0.0871
6	Hydrothermal	0.0623	Mos	0.0760	Challenge	0.0667	Common	0.0686
7	Sulfur	0.0579	Core	0.0706	Eg	0.0662	Carbon	0.0668
8	Process	0.0545	Source	0.0705	Sulfur	0.0659	Involves	0.0583
9	ambient	0.0499	Hours	0.0668	Melt quenching	0.0655	Oxidation	0.0507
10	sodium	0.0464	Hydrothema	0.0663	Scalable	0.0643	Reduction	0.0499

6. How to use LLM about RIMES ?

Classification을 통한 LLM 분류

- 1. The classification model perfectly identified which LLM generated each response about five renewable energy storage materials and their synthesis methods
- 2. This suggests that each LLM produced distinct responses, indicating that they may be suited to different types of use depending on purpose

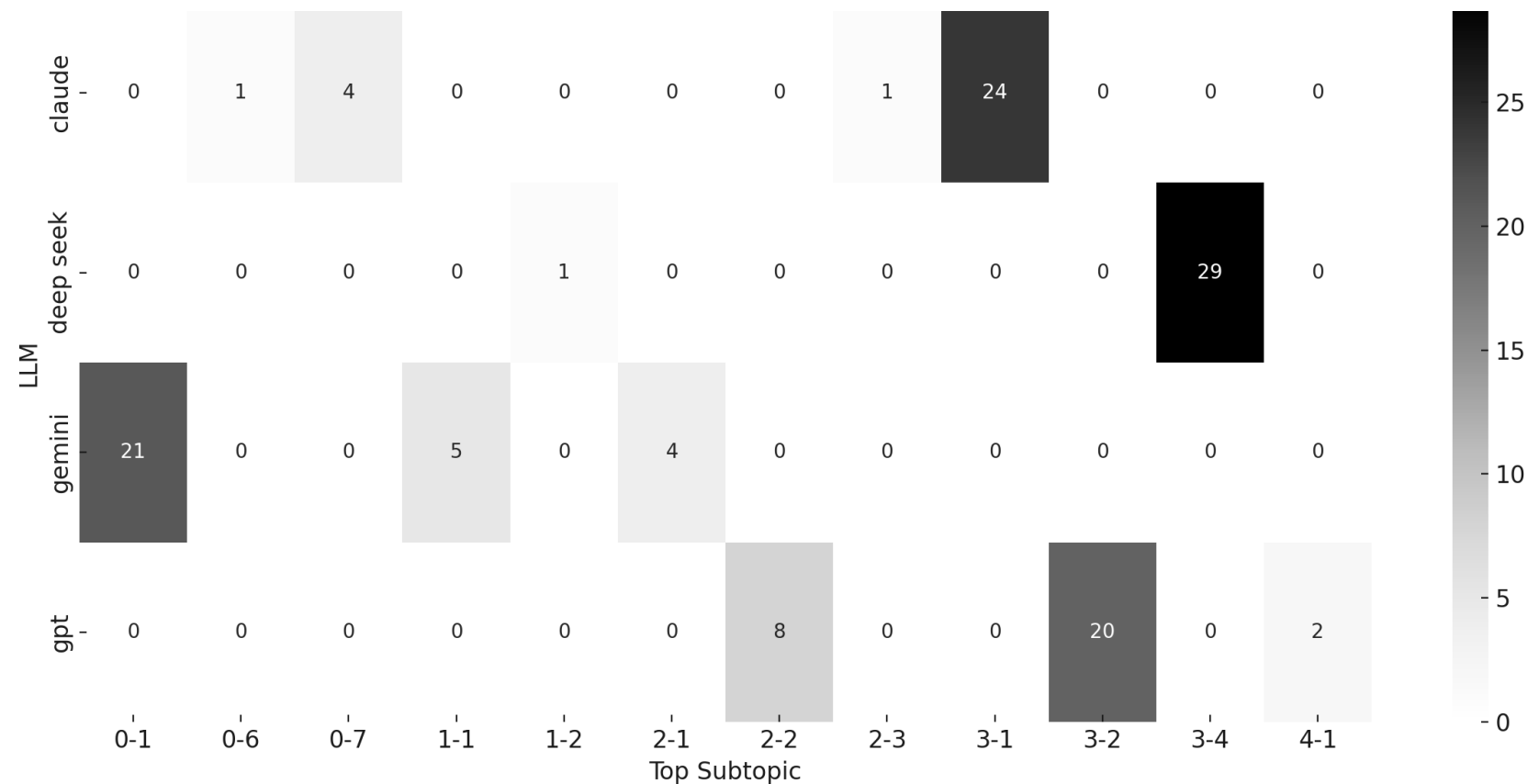
	precision	recall	f1-score	support
claude	1	1	1	12
deep seek	1	1	1	12
gemini	1	1	1	12
gpt	1	1	1	12
accuracy	1			
macro avg	1	1	1	48
weighted avg	1	1	1	48



6. How to use LLM about RIMES ?

Interpreting results with subtopics and probability distribution matching(1)

- 1. This heatmap visualizes the distribution of the subtopics selected as the top topics in the responses generated by each LLM (GPT, Claude, Gemini, and DeepSeek)



6. How to use LLM about RIMES ?

Interpreting results with subtopics and probability distribution matching(2) – Sub topic status

Gemini

0-1 → synthesis (0.021), sulfur (0.021), carbon (0.019), method (0.016), advantage (0.014), vanadium (0.014), principle (0.013), high (0.012), control (0.010), temperature (0.010), reaction (0.010)

1-1 → synthesis (0.022), method (0.021), carbon (0.020), principle (0.020), sulfur (0.019), vanadium (0.016), core (0.015), advantage (0.014), temperatures (0.013), high (0.012), precursors (0.012), temperature (0.012), solution (0.012), reaction (0.011), control (0.011), lfp (0.010)

2-1 → sulfur (0.027), carbon (0.026), synthesis (0.024), method (0.019), vanadium (0.017), advantage (0.016), high (0.014), temperature (0.013),

Claude

3-1 → method (0.023), synthesis (0.022), core (0.020), principle (0.020), methods (0.018), advantage (0.017), carbon (0.016), solid (0.014), state (0.014), high (0.012), vanadium (0.012), oxidation (0.011)

0-7 → method (0.027), principle (0.026), synthesis (0.026), core (0.024), advantage (0.022), methods (0.019), involves (0.015), high (0.015), oxidation (0.013), solid (0.013), state (0.013), carbon (0.013), vanadium (0.013),

0-6 → synthesis (0.020), process (0.020), methods (0.019), carbon (0.018), core (0.018), method (0.017), principle (0.017), state (0.016), solid (0.016), precursors (0.013), reduction (0.012), acid (0.012),

ChatGPT

2-2 → synthesis (0.040), advantage (0.026), high (0.025), method (0.023), methods (0.021), state (0.020), core (0.020), solid (0.018), reaction (0.017), hydrothermal (0.015), vanadium (0.015)

3-2 → synthesis (0.031), advantage (0.020), high (0.019), principle (0.018), method (0.018), core (0.016), methods (0.016), state (0.015), description (0.014), solid (0.014), reaction (0.014)

4-1 → synthesis (0.036), advantage (0.023), high (0.022), principle (0.021), method (0.021), methods (0.019), core (0.019)

DeepSeek

1-2 → synthesis (0.047), advantage (0.039), high (0.032), process (0.032), methods (0.031), solid (0.028), state (0.028), hydrothermal (0.020), method (0.016), sulfur (0.015), chemical (0.014), llzo (0.014), battery (0.014), lfp (0.013)

3-4 → synthesis (0.036), advantage (0.032), process (0.032), high (0.025), methods (0.024), solid (0.022), state (0.021), hydrothermal (0.017), sulfur (0.014), scalable (0.012), llzo (0.011), electrolyte (0.011)

6. How to use LLM about RIMES ?

Interpreting results with subtopics and probability distribution matching(3) – Weight colculation

$$\sum_{d \in D_{\text{LLM}}} \sum_{k \in K_{t_d}} w_{k,t_d}$$

Cumulative weighted sum of keywords by LLM

D_{LLM} - The set of documents generated by that LLM

t_d - Subtopics with the highest probability in Doc d

K_{t_d} - List of keywords in the topic t_d

w_{k,t_d} - Weight assigned to keyword K in topic t_d

Gemini

- Synthesis(0.647), sulfur(0.644), carbon(0.603),
method(0.517), vanadium(0.442), principle(0.373),
high(0.368), temperature(0.322)

ChatGPT

- Synthesis(1.012), advantage(0.645), high(0.624),
method(0.586), core(0.518), state(0.494), solid(0.456),
reaction(0.446)

Claude

- Method(0.706), synthesis(0.680), core(0.620),
principle(0.601), methods(0.550), advantage(0.529),
carbon(0.474), solid(0.422)

DeepSeek

- Synthesis(1.091), advantage(0.967), process(0.960),
high(0.757), methods(0.727), solid(0.666), state(0.637),
hydrothermal(0.513)

6. How to use LLM about RIMES ?

Interpreting results with subtopics and probability distribution matching(4)

Gemini

- Tend to describe material properties (sulfur, carbon, vanadium) and synthesis procedures (synthesis, method) in a balanced manner

Claude

- Suggests that Claude values conceptual understanding of method, principle, and responds strongly to core and structural explanations

ChatGPT

- Response structures that compare and analyze different synthesis methods and their effects, and a broad application of technical explanations are strong

DeepSeek

- Demonstrate that you have created a practical explanation that focuses on the efficiency, applicability, and process-orientation of synthesis methods

7. Conclusion

- First, our research confirms that distinctive linguistic patterns are a measurable and clear differentiator between Human and LLM-generated text, rooted in fundamental differences in semantic diversity and narrative style
- Second, this enables reliable automated classification and provides a crucial tool to address future challenges like “Model Collapse”, ensuring a more transparent and sustainable AI ecosystem
- Third, our results demonstrate that each LLM exhibits consistent and distinct topical preferences, with GPT focusing on descriptive synthesis procedures, Claude emphasizing conceptual principles and temperature conditions, DeepSeek prioritizing scalable and industrial applicability, and Gemini balancing materials and reaction environments.
- Lastly, these weighted linguistic patterns offer a measurable basis for model differentiation and inform strategic deployment in scientific, educational, and industrial domains, contributing to transparent LLM evaluation and robust downstream task alignment

Thank you for your attention

Do you have any question??



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Research Institute for Materials and Energy Sciences